Final Report

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First, we will load the required packages

```
#Loading packages
library(tidyverse)
library(skimr)
library(knitr)
library(ggthemes)
library(ggpubr)
library(patchwork)
library(stringr)
library(lubridate)
library(showtext)
library(flextable)
library(janitor)
library(wordcloud2)
```

I am going to be using the given “Presidential Elections Data: State Level” data set, let’s read it in:

```
#Importing data into R
elections_data <- read_csv("https://raw.githubusercontent.com/dilernia/STA418-518/main/Data/1976-2020-president.csv")
```

```
## Rows: 4287 Columns: 15
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr (6): state, state_po, office, candidate, party_detailed, party_simplified
## dbl (7): year, state_fips, state_cen, state_ic, candidatevotes, totalvotes, ...
## lgl (2): writein, notes
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Let’s skim the dataset

```
skim(elections_data)
```

Data summary

Name	elections_data
Number of rows	4287
Number of columns	15
_______________________	
Column type frequency:	
character	6
logical	2
numeric	7
________________________	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
state	0	1.00	4	20	0	51	0
state_po	0	1.00	2	2	0	51	0
office	0	1.00	12	12	0	1	0
candidate	287	0.93	4	38	0	270	0
party_detailed	456	0.89	3	40	0	172	0
party_simplified	0	1.00	5	11	0	4	0

Variable type: logical

skim_variable	n_missing	complete_rate	mean	count
writein	3	1	0.11	FAL: 3807, TRU: 477
notes	4287	0	NaN	:

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1	1999.08	14.22	1976	1988	2000	2012.0	2020	▇▆▆▆▇
state_fips	0	1	28.62	15.62	1	16	28	41.0	56	▇▇▇▇▇
state_cen	0	1	53.67	26.03	11	33	53	81.0	95	▆▆▇▃▇
state_ic	0	1	39.75	22.77	1	22	42	61.0	82	▇▆▇▇▅
candidatevotes	0	1	311907.59	764801.10	0	1177	7499	199241.5	11110250	▇▁▁▁▁
totalvotes	0	1	2366924.15	2465008.31	123574	652274	1569180	3033118.0	17500881	▇▂▁▁▁
version	0	1	20210113.00	0.00	20210113	20210113	20210113	20210113.0	20210113	▁▁▇▁▁

After importing the data set and using the skim function, we can analyze the data. The dataset contains 4,287 rows and 15 columns, representing 15 variables. More in detail, we have 6 character variables, 2 logical ones and 7 numeric. Looking at the m_missing column, we learn that all the numeric and 4 of the character variables have no missing values. Candidate has 287 missing values, so its complete rate is 0.89 and Party_detailed has 456 missing values, which gives it a 0.89 complete rate. This is not a big deal, since the candidates and parties missing all got a really low number of votes, so they won't be used. We can also point out that the variable Office has only one unique value, it been "US PRESIDENT", since all candidates were running for the same office. Therefore, this variable won't be of any use, so we could remove it. When looking at the logic ones on the other hand, we find that in the writein variable we are missing 3 values, which is not substantial since the complete rate is 0.9993002. For the other logic variable, notes, we can see that all of the values are missing. We will later get rid of this column since it does not provide any information.

We will also explore the data with the glimpse function

```
glimpse(elections_data)
```

```
## Rows: 4,287
## Columns: 15
## $ year             <dbl> 1976, 1976, 1976, 1976, 1976, 1976, 1976, 1976, 1976,…
## $ state            <chr> "ALABAMA", "ALABAMA", "ALABAMA", "ALABAMA", "ALABAMA"…
## $ state_po         <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AK", "AK",…
## $ state_fips       <dbl> 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 4, 4, 4, 4, 4, 4, 4,…
## $ state_cen        <dbl> 63, 63, 63, 63, 63, 63, 63, 94, 94, 94, 94, 86, 86, 8…
## $ state_ic         <dbl> 41, 41, 41, 41, 41, 41, 41, 81, 81, 81, 81, 61, 61, 6…
## $ office           <chr> "US PRESIDENT", "US PRESIDENT", "US PRESIDENT", "US P…
## $ candidate        <chr> "CARTER, JIMMY", "FORD, GERALD", "MADDOX, LESTER", "B…
## $ party_detailed   <chr> "DEMOCRAT", "REPUBLICAN", "AMERICAN INDEPENDENT PARTY…
## $ writein          <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, TRUE, FALSE…
## $ candidatevotes   <dbl> 659170, 504070, 9198, 6669, 1954, 1481, 308, 71555, 4…
## $ totalvotes       <dbl> 1182850, 1182850, 1182850, 1182850, 1182850, 1182850,…
## $ version          <dbl> 20210113, 20210113, 20210113, 20210113, 20210113, 202…
## $ notes            <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
## $ party_simplified <chr> "DEMOCRAT", "REPUBLICAN", "OTHER", "OTHER", "OTHER", …
```

After using the glimpse function, we can now see with more detail what type each variable is and take a look at the first rows. Now we can confirm the patterns I explained above when using the skim function. We can see how the office variable always has the same value, or how the notes column is empty.

1. Data Dictionary

We are going to print the main information explained at the top in a data dictionary:

```
#Creating data dictionary
dataDictionary1 <- tibble(Variable = colnames(elections_data),
                          Description = c("Election year",
                                          "State name",
                                          "State postal code abbreviation",
                                          "State FIPS code",
                                          "State census code",
                                          "State Inter-university Consortium for Political and Social Research code",
                                          "Name of the public office to which the candidate is seeking election",
                                          "Candidate name",
                                          "Full name of the candidate's political party",
                                          "Candidate is a write-in (TRUE or FALSE)",
                                          "Number of votes for candidate",
                                          "Total votes cast in the election for the state",
                                          "Version",
                                          "Notes",
                                          "Just the major parties, with others marked as other"),
                          Type = map_chr(elections_data, .f = function(x){typeof(x)[1]}),
                          Class = map_chr(elections_data, .f = function(x){class(x)[1]}))

# Printing nicely in R Markdown
flextable::flextable(dataDictionary1, cwidth = 2)
```

Variable	Description	Type	Class
year	Election year	double	numeric
state	State name	character	character
state_po	State postal code abbreviation	character	character
state_fips	State FIPS code	double	numeric
state_cen	State census code	double	numeric
state_ic	State Inter-university Consortium for Political and Social Research code	double	numeric
office	Name of the public office to which the candidate is seeking election	character	character
candidate	Candidate name	character	character
party_detailed	Full name of the candidate's political party	character	character
writein	Candidate is a write-in (TRUE or FALSE)	logical	logical
candidatevotes	Number of votes for candidate	double	numeric
totalvotes	Total votes cast in the election for the state	double	numeric
version	Version	double	numeric
notes	Notes	logical	logical
party_simplified	Just the major parties, with others marked as other	character	character

2. Data Cleaning

a. Merging Data Sets

It can be really interesting to look for patterns in the different states where democrats or republicans where the most voting party. Does income, race, education, gender or any other factors have anything to do with the number of votes each party is getting? To find this out I will be combining the previously explained elections data set with a new one, which has census data from every state from the years 2008 trough 2021.

First, let's read the data set:

```
#Importing data
census_data <- read_csv("https://raw.githubusercontent.com/dilernia/STA418-518/main/Data/census_data_state_2008-2021.csv")
```

```
## Rows: 676 Columns: 26
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr  (2): geoid, county_state
## dbl (24): year, population, median_income, median_monthly_rent_cost, median_...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

First, let's skim the data:

```
skim(census_data)
```

Data summary

Name	census_data
Number of rows	676
Number of columns	26
_______________________	
Column type frequency:	
character	2
numeric	24
________________________	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
geoid	0	1	2	2	0	52	0
county_state	0	1	4	20	0	52	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1	2014.08	3.88	2008.00	2011.00	2014.00	2017.00	2021.00	
population	0	1	6190132.93	6978987.03	532668.00	1786156.75	4257700.00	6972359.00	39557045.00	
median_income	0	1	55793.43	12152.82	18314.00	47512.50	54740.00	63255.75	92266.00	
median_monthly_rent_cost	0	1	900.19	230.04	419.00	735.75	847.50	1024.50	1774.00	
median_monthly_home_cost	0	1	1084.63	360.03	241.00	833.50	987.50	1314.25	2215.00	
prop_female	0	1	0.51	0.01	0.47	0.50	0.51	0.51	0.53	
prop_male	0	1	0.49	0.01	0.47	0.49	0.49	0.50	0.53	
prop_white	0	1	0.76	0.14	0.22	0.68	0.78	0.86	0.96	
prop_black	0	1	0.11	0.11	0.00	0.03	0.07	0.16	0.53	
prop_native	0	1	0.02	0.03	0.00	0.00	0.00	0.01	0.16	
prop_asian	0	1	0.04	0.05	0.00	0.01	0.02	0.04	0.39	
prop_hawaiin_islander	0	1	0.00	0.01	0.00	0.00	0.00	0.00	0.11	
prop_other_race	0	1	0.03	0.03	0.00	0.01	0.02	0.05	0.30	
prop_multi_racial	0	1	0.04	0.04	0.01	0.02	0.03	0.03	0.36	
prop_highschool	0	1	0.24	0.04	0.12	0.22	0.24	0.27	0.35	
prop_GED	0	1	0.04	0.01	0.02	0.04	0.04	0.05	0.07	
prop_some_college	0	1	0.06	0.01	0.01	0.06	0.06	0.07	0.09	
prop_college_no_degree	0	1	0.15	0.02	0.09	0.13	0.15	0.16	0.22	
prop_associates	0	1	0.08	0.02	0.03	0.07	0.08	0.09	0.15	
prop_bachelors	0	1	0.19	0.03	0.10	0.17	0.19	0.21	0.27	
prop_masters	0	1	0.08	0.03	0.04	0.06	0.07	0.09	0.23	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
prop_professional	0	1	0.02	0.01	0.01	0.02	0.02	0.02	0.10	▇▁▁▁▁
prop_doctoral	0	1	0.01	0.01	0.01	0.01	0.01	0.02	0.05	▇▂▁▁▁
prop_poverty	0	1	0.14	0.05	0.07	0.11	0.14	0.16	0.46	▇▆▁▁▁

After skimming the data, we can see we have 2 character variables and 24 numeric ones. All of the 26 variables have a complete rate of 1, meaning there are no missing values, as we can see in the n_missing row.

Let's take a glimpse at this data set to see which variables could be useful:

```
glimpse(census_data)
```

```
## Rows: 676
## Columns: 26
## $ geoid                    <chr> "01", "02", "04", "05", "06", "08", "09", "10…
## $ county_state             <chr> "Alabama", "Alaska", "Arizona", "Arkansas", "…
## $ year                     <dbl> 2008, 2008, 2008, 2008, 2008, 2008, 2008, 200…
## $ population               <dbl> 4661900, 686293, 6500180, 2855390, 36756666, …
## $ median_income            <dbl> 42666, 68460, 50958, 38815, 61021, 56993, 685…
## $ median_monthly_rent_cost <dbl> 631, 949, 866, 606, 1135, 848, 970, 917, 1011…
## $ median_monthly_home_cost <dbl> 742, 1393, 1177, 660, 1919, 1364, 1722, 1209,…
## $ prop_female              <dbl> 0.5154499, 0.4789995, 0.4986763, 0.5102035, 0…
## $ prop_male                <dbl> 0.4845501, 0.5210005, 0.5013237, 0.4897965, 0…
## $ prop_white               <dbl> 0.7028360, 0.6911290, 0.8005912, 0.7871510, 0…
## $ prop_black               <dbl> 0.261833802, 0.036257109, 0.036269457, 0.1550…
## $ prop_native              <dbl> 0.005140822, 0.127259057, 0.044114009, 0.0052…
## $ prop_asian               <dbl> 0.009768335, 0.046373779, 0.023755650, 0.0103…
## $ prop_hawaiin_islander    <dbl> 2.417469e-04, 5.484538e-03, 1.436883e-03, 6.7…
## $ prop_other_race          <dbl> 0.007177117, 0.012462607, 0.068086884, 0.0218…
## $ prop_multi_racial        <dbl> 0.013002209, 0.081033902, 0.025745902, 0.0197…
## $ prop_highschool          <dbl> 0.2604227, 0.2183097, 0.2106347, 0.2958009, 0…
## $ prop_GED                 <dbl> 0.05599385, 0.04658968, 0.03935913, 0.0582695…
## $ prop_some_college        <dbl> 0.05957806, 0.08116479, 0.07719126, 0.0687879…
## $ prop_college_no_degree   <dbl> 0.1546491, 0.2168623, 0.1816943, 0.1534883, 0…
## $ prop_associates          <dbl> 0.06831511, 0.08030577, 0.07837909, 0.0558112…
## $ prop_bachelors           <dbl> 0.1427926, 0.1758716, 0.1590928, 0.1251352, 0…
## $ prop_masters             <dbl> 0.05542474, 0.06465035, 0.06474227, 0.0433444…
## $ prop_professional        <dbl> 0.01338380, 0.01743700, 0.01623786, 0.0121880…
## $ prop_doctoral            <dbl> 0.008333932, 0.015125876, 0.010927963, 0.0072…
## $ prop_poverty             <dbl> 0.15695141, 0.08420041, 0.14718370, 0.1732244…
```

Now, let's create a data dictionary for the census data set:

```r
#Creating data dictionary
dataDictionary2 <- tibble(Variable = colnames(census_data),
                          Description = c("Geographic region ID with the first 2 digits being the state Federal Information P
rocessing Standard (FIPS) code and the last 3 digits the county FIPS code",
                                          "Geographic region",
                                          "Year",
                                          "Population",
                                          "Median income in dollars",
                                          "Median monthly rent costs for renters in dollars",
                                          "Median monthly housing costs for homeowners in dollars",
                                          "Proportion of people who are female",
                                          "Proportion of people who are male",
                                          "Proportion of people who are white alone",
                                          "Proportion of people who are black or African American alone",
                                          "Proportion of people who are American Indian and Alaska Native alone",
                                          "Proportion of people who are Asian alone",
                                          "Proportion of people who are Native Hawaiian and Other Pacific Islander alone",
                                          "Proportion of people who are some other race alone",
                                          "Proportion of people who are two or more races",
                                          "Proportion of people 25 and older whose highest education level is high school",
                                          "Proportion of people 25 and older whose highest education level is a GED",
                                          "Proportion of people 25 and older whose highest education level is some, but less
than 1 year of college",
                                          "Proportion of people 25 and older whose highest education level is greater than 1
year of college but no degree",
                                          "Proportion of people 25 and older whose highest education level is an Associate's
degree",
                                          "Proportion of people 25 and older whose highest education level is a Bachelor's de
gree",
                                          "Proportion of people 25 and older whose highest education level is a Master's degr
ee",
                                          "Proportion of people 25 and older whose highest education level is a Professional
degree",
                                          "Proportion of people 25 and older whose highest education level is a Doctoral degr
ee",
                                          "Proportion of people 25 and older living in poverty, defined by the Census Bureau
as having an income below the poverty threshold for their family size"),
                          Type = map_chr(census_data, .f = function(x){typeof(x)[1]}),
                          Class = map_chr(census_data, .f = function(x){class(x)[1]}))

# Printing nicely in R Markdown
flextable::flextable(dataDictionary2, cwidth = 2)
```

Variable	Description	Type	Class
geoid	Geographic region ID with the first 2 digits being the state Federal Information Processing Standard (FIPS) code and the last 3 digits the county FIPS code	character	character
county_state	Geographic region	character	character
year	Year	double	numeric
population	Population	double	numeric
median_income	Median income in dollars	double	numeric
median_monthly_rent_cost	Median monthly rent costs for renters in dollars	double	numeric
median_monthly_home_cost	Median monthly housing costs for homeowners in dollars	double	numeric
prop_female	Proportion of people who are female	double	numeric
prop_male	Proportion of people who are male	double	numeric
prop_white	Proportion of people who are white alone	double	numeric
prop_black	Proportion of people who are black or African American alone	double	numeric
prop_native	Proportion of people who are American Indian and Alaska Native alone	double	numeric

Variable	Description	Type	Class
prop_asian	Proportion of people who are Asian alone	double	numeric
prop_hawaiin_islander	Proportion of people who are Native Hawaiian and Other Pacific Islander alone	double	numeric
prop_other_race	Proportion of people who are some other race alone	double	numeric
prop_multi_racial	Proportion of people who are two or more races	double	numeric
prop_highschool	Proportion of people 25 and older whose highest education level is high school	double	numeric
prop_GED	Proportion of people 25 and older whose highest education level is a GED	double	numeric
prop_some_college	Proportion of people 25 and older whose highest education level is some, but less than 1 year of college	double	numeric
prop_college_no_degree	Proportion of people 25 and older whose highest education level is greater than 1 year of college but no degree	double	numeric
prop_associates	Proportion of people 25 and older whose highest education level is an Associate's degree	double	numeric
prop_bachelors	Proportion of people 25 and older whose highest education level is a Bachelor's degree	double	numeric
prop_masters	Proportion of people 25 and older whose highest education level is a Master's degree	double	numeric
prop_professional	Proportion of people 25 and older whose highest education level is a Professional degree	double	numeric
prop_doctoral	Proportion of people 25 and older whose highest education level is a Doctoral degree	double	numeric
prop_poverty	Proportion of people 25 and older living in poverty, defined by the Census Bureau as having an income below the poverty threshold for their family size	double	numeric

Before merging the data sets, we need to get the elections data set ready, since for every state-year combination, there is more than one entry, since we have one for every candidate in each state-year combination. On the census data set, we only have each state-year combination once.

First we will drop the variables we don't need:

```
#Dropping variables
elections_new <- elections_data |>
  dplyr::select(-office, -notes, -writein, -version, -state_po, -state_fips, -state_cen, -state_ic, -totalvotes)
```

We are going to add a variable called total_votes_us that measures the total votes in the country of each candidate of the main parties each year

```
#Adding new variable for total votes
elections_new <- elections_new |>
  group_by(year, candidate, party_simplified) |>
  mutate(total_votes_us = sum(candidatevotes, na.rm = TRUE)) |>
  ungroup()
```

After this, we are going to keep only one row for each state-year combination. More specifically, we will keep the row with the candidate with more votes for every state-year combination. In other words, we will keep the row of the candidate who won each state.

```
#Keep only the row of the winning party for every state-year combination
elections_summarized <- elections_new |>
  group_by(state, year) |>
  slice_max(order_by = candidatevotes, n = 1) |>
  ungroup()

#Let's sort it by year, not state
elections_summarized <- elections_summarized |>
  arrange(year, state)

#Finally, let's take a glimpse at the new data set to check if it worked
glimpse(elections_summarized)
```

```
## Rows: 612
## Columns: 7
## $ year             <dbl> 1976, 1976, 1976, 1976, 1976, 1976, 1976, 1976, 1976,…
## $ state            <chr> "ALABAMA", "ALASKA", "ARIZONA", "ARKANSAS", "CALIFORN…
## $ candidate        <chr> "CARTER, JIMMY", "FORD, GERALD", "FORD, GERALD", "CAR…
## $ party_detailed   <chr> "DEMOCRAT", "REPUBLICAN", "REPUBLICAN", "DEMOCRAT", "…
## $ candidatevotes   <dbl> 659170, 71555, 418642, 498604, 3882244, 584278, 71926…
## $ party_simplified <chr> "DEMOCRAT", "REPUBLICAN", "REPUBLICAN", "DEMOCRAT", "…
## $ total_votes_us   <dbl> 40680446, 38870893, 38870893, 40680446, 38870893, 388…
```

Since on the census data set each state is written as a title (Michigan) and on the elections data all the letters are uppercase (MICHIGAN), we will change the format on this last data set so that the merge is possible.

```
#Change state format
elections_summarized <- elections_summarized |>
  mutate(state = str_to_title(state))
```

Let's get back to the census data set before merging. There is no data for the year 2020, due to COVID. Since this is an election year, it could be really useful for the analysis. Therefore, I am going to change the year of the 2021 data to 2020. Since it is from the following year it won't differ much from the 2020 data. This way, we will have demographic data of one more election year.

```
#Changing 2021 to 2020
census_data <- census_data |>
  mutate(year = if_else(year == 2021, 2020, year))
```

Before merging the data sets, we need to keep a few things in mind. The elections data set starts in 1976 and the census in 2008. Therefore, we can't use a left join since the data from 1976 to 2007 will have no values for the census variables. Also, since elections are every 4 years, we can't use a right join because for the non-election years, the elections variables would be empty, Therefore, the appropriate join is an inner join.

```
# Merge elections_summarized with census_data
merged_data <- elections_summarized |>
  inner_join(census_data, by = c("state" = "county_state", "year" = "year"))

# Verify the merged data
glimpse(merged_data)
```

```
## Rows: 200
## Columns: 31
## $ year                     <dbl> 2008, 2008, 2008, 2008, 2008, 2008, 2008, 200…
## $ state                    <chr> "Alabama", "Alaska", "Arizona", "Arkansas", "…
## $ candidate                <chr> "MCCAIN, JOHN", "MCCAIN, JOHN", "MCCAIN, JOHN…
## $ party_detailed           <chr> "REPUBLICAN", "REPUBLICAN", "REPUBLICAN", "RE…
## $ candidatevotes           <dbl> 1266546, 193841, 1230111, 638017, 8274473, 12…
## $ party_simplified         <chr> "REPUBLICAN", "REPUBLICAN", "REPUBLICAN", "RE…
## $ total_votes_us           <dbl> 59613835, 59613835, 59613835, 59613835, 69338…
## $ geoid                    <chr> "01", "02", "04", "05", "06", "08", "09", "10…
## $ population               <dbl> 4661900, 686293, 6500180, 2855390, 36756666, …
## $ median_income            <dbl> 42666, 68460, 50958, 38815, 61021, 56993, 685…
## $ median_monthly_rent_cost <dbl> 631, 949, 866, 606, 1135, 848, 970, 917, 947,…
## $ median_monthly_home_cost <dbl> 742, 1393, 1177, 660, 1919, 1364, 1722, 1209,…
## $ prop_female              <dbl> 0.5154499, 0.4789995, 0.4986763, 0.5102035, 0…
## $ prop_male                <dbl> 0.4845501, 0.5210005, 0.5013237, 0.4897965, 0…
## $ prop_white               <dbl> 0.7028360, 0.6911290, 0.8005912, 0.7871510, 0…
## $ prop_black               <dbl> 0.261833802, 0.036257109, 0.036269457, 0.1550…
## $ prop_native              <dbl> 0.005140822, 0.127259057, 0.044114009, 0.0052…
## $ prop_asian               <dbl> 0.009768335, 0.046373779, 0.023755650, 0.0103…
## $ prop_hawaiin_islander    <dbl> 2.417469e-04, 5.484538e-03, 1.436883e-03, 6.7…
## $ prop_other_race          <dbl> 0.007177117, 0.012462607, 0.068086884, 0.0218…
## $ prop_multi_racial        <dbl> 0.013002209, 0.081033902, 0.025745902, 0.0197…
## $ prop_highschool          <dbl> 0.2604227, 0.2183097, 0.2106347, 0.2958009, 0…
## $ prop_GED                 <dbl> 0.05599385, 0.04658968, 0.03935913, 0.0582695…
## $ prop_some_college        <dbl> 0.05957806, 0.08116479, 0.07719126, 0.0687879…
## $ prop_college_no_degree   <dbl> 0.1546491, 0.2168623, 0.1816943, 0.1534883, 0…
## $ prop_associates          <dbl> 0.06831511, 0.08030577, 0.07837909, 0.0558112…
## $ prop_bachelors           <dbl> 0.1427926, 0.1758716, 0.1590928, 0.1251352, 0…
## $ prop_masters             <dbl> 0.05542474, 0.06465035, 0.06474227, 0.0433444…
## $ prop_professional        <dbl> 0.01338380, 0.01743700, 0.01623786, 0.0121880…
## $ prop_doctoral            <dbl> 0.008333932, 0.015125876, 0.010927963, 0.0072…
## $ prop_poverty             <dbl> 0.15695141, 0.08420041, 0.14718370, 0.1732244…
```

```
skim(merged_data)
```

Data summary

Name	merged_data
Number of rows	200
Number of columns	31
_______________________	
Column type frequency:	
character	5
numeric	26
________________________	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
state	0	1	4	14	0	50	0
candidate	0	1	12	19	0	6	0
party_detailed	0	1	8	23	0	3	0
party_simplified	0	1	8	10	0	2	0
geoid	0	1	2	2	0	50	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
year	0	1	2014.00	4.48	2008.00	2011.00	2014.00	2017.00	2020.00
candidatevotes	0	1	1545144.54	1610451.75	163387.00	432534.25	1150268.50	1974233.25	11110250.00
total_votes_us	0	1	67345512.99	6998635.89	59379447.00	62692670.00	65752017.00	70558171.00	81268908.00
population	0	1	6352098.26	7065435.80	532668.00	1826943.50	4458387.00	7279237.00	39250017.00

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
median_income	0	1	57572.58	11703.40	37095.00	48580.00	55929.50	65758.00	90203.00
median_monthly_rent_cost	0	1	912.58	235.77	528.00	741.00	861.50	1035.25	1774.00
median_monthly_home_cost	0	1	1097.20	332.45	494.00	834.00	1004.00	1319.50	1919.00
prop_female	0	1	0.51	0.01	0.47	0.50	0.51	0.51	0.52
prop_male	0	1	0.49	0.01	0.48	0.49	0.49	0.50	0.53
prop_white	0	1	0.75	0.14	0.22	0.68	0.78	0.85	0.96
prop_black	0	1	0.10	0.09	0.00	0.03	0.07	0.15	0.38
prop_native	0	1	0.02	0.03	0.00	0.00	0.01	0.01	0.15
prop_asian	0	1	0.04	0.06	0.01	0.01	0.02	0.04	0.38
prop_hawaiin_islander	0	1	0.00	0.01	0.00	0.00	0.00	0.00	0.10
prop_other_race	0	1	0.03	0.03	0.00	0.01	0.02	0.05	0.19
prop_multi_racial	0	1	0.05	0.05	0.01	0.02	0.03	0.06	0.28
prop_highschool	0	1	0.24	0.04	0.16	0.22	0.24	0.27	0.34
prop_GED	0	1	0.04	0.01	0.02	0.04	0.04	0.05	0.07
prop_some_college	0	1	0.07	0.01	0.05	0.06	0.06	0.07	0.09
prop_college_no_degree	0	1	0.15	0.02	0.09	0.13	0.15	0.16	0.22
prop_associates	0	1	0.09	0.02	0.05	0.08	0.08	0.09	0.14
prop_bachelors	0	1	0.19	0.03	0.10	0.17	0.19	0.21	0.27
prop_masters	0	1	0.08	0.02	0.04	0.06	0.08	0.09	0.15
prop_professional	0	1	0.02	0.00	0.01	0.02	0.02	0.02	0.03
prop_doctoral	0	1	0.01	0.00	0.01	0.01	0.01	0.02	0.03
prop_poverty	0	1	0.13	0.03	0.07	0.11	0.13	0.16	0.24

After reviewing the newly created data set, we can see there are no missing values in any of the variables. We now have 31 total variables and 200 rows. The census data set had data on the 50 states, and since 4 election years appear on it, we have 200 rows of data, from the years 2008, 2012, 2016 and 2020.

Let's create another data dictionary for the merged data set, having in mind all the modifications done before the merging:

```
#Creating data dictionary
dataDictionary3 <- tibble(Variable = colnames(merged_data),
                         Description = c("Election year",
                                         "State name",
                                         "Candidate name",
                                         "Full name of the candidate's political party",
                                         "Number of votes for candidate",
                                         "Just the major parties, with others marked as other",
                                         "Total votes in the election that year in the US",
                                         "Geographic region ID with the first 2 digits being the state Federal Information P
rocessing Standard (FIPS) code and the last 3 digits the county FIPS code",
                                         "Population",
                                         "Median income in dollars",
                                         "Median monthly rent costs for renters in dollars",
                                         "Median monthly housing costs for homeowners in dollars",
                                         "Proportion of people who are female",
                                         "Proportion of people who are male",
                                         "Proportion of people who are white alone",
                                         "Proportion of people who are black or African American alone",
                                         "Proportion of people who are American Indian and Alaska Native alone",
                                         "Proportion of people who are Asian alone",
                                         "Proportion of people who are Native Hawaiian and Other Pacific Islander alone",
                                         "Proportion of people who are some other race alone",
                                         "Proportion of people who are two or more races",
                                         "Proportion of people 25 and older whose highest education level is high school",
                                         "Proportion of people 25 and older whose highest education level is a GED",
                                         "Proportion of people 25 and older whose highest education level is some, but less
than 1 year of college",
                                         "Proportion of people 25 and older whose highest education level is greater than 1
year of college but no degree",
                                         "Proportion of people 25 and older whose highest education level is an Associate's
degree",
                                         "Proportion of people 25 and older whose highest education level is a Bachelor's de
gree",
                                         "Proportion of people 25 and older whose highest education level is a Master's degr
ee",
                                         "Proportion of people 25 and older whose highest education level is a Professional
degree",
                                         "Proportion of people 25 and older whose highest education level is a Doctoral degr
ee",
                                         "Proportion of people 25 and older living in poverty, defined by the Census Bureau
as having an income below the poverty threshold for their family size"),
                         Type = map_chr(merged_data, .f = function(x){typeof(x)[1]}),
                         Class = map_chr(merged_data, .f = function(x){class(x)[1]}))

# Printing nicely in R Markdown
flextable::flextable(dataDictionary3, cwidth = 2)
```

Variable	Description	Type	Class
year	Election year	double	numeric
state	State name	character	character
candidate	Candidate name	character	character
party_detailed	Full name of the candidate's political party	character	character
candidatevotes	Number of votes for candidate	double	numeric
party_simplified	Just the major parties, with others marked as other	character	character
total_votes_us	Total votes in the election that year in the US	double	numeric
geoid	Geographic region ID with the first 2 digits being the state Federal Information Processing Standard (FIPS) code and the last 3 digits the county FIPS code	character	character
population	Population	double	numeric
median_income	Median income in dollars	double	numeric
median_monthly_rent_cost	Median monthly rent costs for renters in dollars	double	numeric
median_monthly_home_cost	Median monthly housing costs for homeowners in	double	numeric

Variable	Description	Type	Class
	dollars		
prop_female	Proportion of people who are female	double	numeric
prop_male	Proportion of people who are male	double	numeric
prop_white	Proportion of people who are white alone	double	numeric
prop_black	Proportion of people who are black or African American alone	double	numeric
prop_native	Proportion of people who are American Indian and Alaska Native alone	double	numeric
prop_asian	Proportion of people who are Asian alone	double	numeric
prop_hawaiin_islander	Proportion of people who are Native Hawaiian and Other Pacific Islander alone	double	numeric
prop_other_race	Proportion of people who are some other race alone	double	numeric
prop_multi_racial	Proportion of people who are two or more races	double	numeric
prop_highschool	Proportion of people 25 and older whose highest education level is high school	double	numeric
prop_GED	Proportion of people 25 and older whose highest education level is a GED	double	numeric
prop_some_college	Proportion of people 25 and older whose highest education level is some, but less than 1 year of college	double	numeric
prop_college_no_degree	Proportion of people 25 and older whose highest education level is greater than 1 year of college but no degree	double	numeric
prop_associates	Proportion of people 25 and older whose highest education level is an Associate's degree	double	numeric
prop_bachelors	Proportion of people 25 and older whose highest education level is a Bachelor's degree	double	numeric
prop_masters	Proportion of people 25 and older whose highest education level is a Master's degree	double	numeric
prop_professional	Proportion of people 25 and older whose highest education level is a Professional degree	double	numeric
prop_doctoral	Proportion of people 25 and older whose highest education level is a Doctoral degree	double	numeric
prop_poverty	Proportion of people 25 and older living in poverty, defined by the Census Bureau as having an income below the poverty threshold for their family size	double	numeric

b. String Manipulation

There are 50 states in the US, as stated before. All these states have lots of different names, and, some of them include meaningful words in their names. For instance, some States start with New (New York) and others contain directional words (North Carolina). We will now see exactly how many of these states there are.

```
#Detecting the pattern with str_detect() function
merged_data <- merged_data |>
  mutate(
    is_new_state = str_detect(state, "^New"),
    is_directional_state = str_detect(state, "North|South|West|East"))

#Only unique values
unique_new_states <- merged_data |>
  filter(is_new_state == TRUE) |>
  distinct(state)
unique_directional_states <- merged_data |>
  filter(is_directional_state == TRUE) |>
  distinct(state)

#Displaying the results

unique_states_combined <- merged_data |>
  mutate(
    state_type = case_when(
      str_detect(state, "^New") ~ "New",
      str_detect(state, "North|South|West|East") ~ "Directional",
      TRUE ~ NA_character_)) |>
  filter(!is.na(state_type)) |>
  distinct(state, state_type)

unique_states_combined |>
  flextable() |>
  set_caption("States with New or Directional words") |>
  theme_vanilla() |>
  autofit()
```

States with New or Directional words

state	state_type
New Hampshire	New
New Jersey	New
New Mexico	New
New York	New
North Carolina	Directional
North Dakota	Directional
South Carolina	Directional
South Dakota	Directional
West Virginia	Directional

As we can see displayed in the above table, there are a total of 4 states starting with the word “New” and 5 starting with directional words.

Now, to improve the visualizations of the tables and graphs that will later be shown, I will update multiple variables of the merged data set as well as from the elections one, since both of them will be used.I will change the variables candidate, party_detailed and party_simplified to a title format:

```
#For the merged data set
merged_data <- merged_data |>
  mutate(
    candidate = str_to_title(candidate),
    party_detailed = str_to_title(party_detailed),
    party_simplified = str_to_title(party_simplified))

#For the elections data set
elections_new <- elections_new |>
  mutate(
    candidate = str_to_title(candidate),
    party_detailed = str_to_title(party_detailed),
    party_simplified = str_to_title(party_simplified))
```

Now all the entries for these variables will have a title format. REPUBLICAN is now Republican

Since the candidates first and last names are in the same column, with a format "Last Name, First Name", we are going to split them and create two new columns, first_name and last_name so we can use them separately. If a last name or first letter of the last name appears in the original name, it will be deleted. In case we need it later, we will apply these changes to both data sets.

```
#For the merged dataset
merged_data <- merged_data |>
  mutate(last_name = str_split(candidate, ",", simplify = TRUE)[, 1],
first_and_middle_name = str_trim(str_split(candidate, ",", simplify = TRUE)[, 2]),
first_name = str_split(first_and_middle_name, " ", simplify = TRUE)[, 1]) |>
  select(-first_and_middle_name)

#For the elections dataset
elections_new <- elections_new |>
  mutate(last_name = str_split(candidate, ",", simplify = TRUE)[, 1],
first_and_middle_name = str_trim(str_split(candidate, ",", simplify = TRUE)[, 2]),
first_name = str_split(first_and_middle_name, " ", simplify = TRUE)[, 1]) |>
  select(-first_and_middle_name)
```

Now two new rows have been created. For example, if the original name was "Obama, Barack H.", now it also appears divided into two columns as "Obama" and "Barack".

3. Exploratory Data Analysis

a. Tables of Summary Statistics

Table for candidate votes grouped by party

For this first table, we are going to see the count and some important statistics of the main political parties in the country.

```
#creating table
candidate_by_party <- elections_new |> group_by(party_simplified) |>
summarize(
  count = n(),
  total_votes = sum(candidatevotes, na.rm = TRUE),
  max_votes = max(candidatevotes, na.rm = TRUE),
  mean_votes = mean(candidatevotes, na.rm = TRUE),
  median_votes = median(candidatevotes, na.rm = TRUE)) |>
  arrange(desc(total_votes))

#Changing col names
candidate_by_party <- candidate_by_party |>
  rename(
    "Party" = party_simplified,
    "Total states featured in" = count,
    "Total votes" = total_votes,
    "Max votes" = max_votes,
    "Mean votes" = mean_votes,
    "Median votes" = median_votes)

#Giving format to table
candidate_by_party |>
  flextable() |>
  set_caption(caption = "Table 1: Summary statistics of votes by party for the elections 1976-2020") |>
  theme_vanilla() |>
  autofit()
```

Table 1: Summary statistics of votes by party for the elections 1976-2020

Party	Total states featured in	Total votes	Max votes	Mean votes	Median votes
Democrat	615	639,242,184	11,110,250	1,039,418.19	649,641
Republican	613	632,502,462	6,006,429	1,031,814.78	718,058
Other	2,524	54,657,973	2,296,006	21,655.30	1,933
Libertarian	535	10,745,232	478,500	20,084.55	7,936

Frequency table of more voted candidates over the years

For this frequency table, we are going to show the candidates with the more total votes in a specific year, which year it is and the votes. We will sort them descending by the number of votes and show the highest 10. For this table, we will also be using the original data set of elections.

```
#Creating table
Most_votes <- elections_new |>
  group_by(year, candidate, party_simplified) |>
  distinct(total_votes_us, .keep_all = TRUE) |>
  ungroup() |>
  select(candidate, year, party_simplified, total_votes_us) |>
  arrange(desc(total_votes_us)) |>
  slice_head(n = 10)

#Changing col names

Most_votes <- Most_votes |>
  rename(
    "Candidate" = candidate,
    "Year" = year,
    "Party" = party_simplified,
    "Total votes" = total_votes_us)

#Giving format to table
Most_votes |>
  flextable() |>
  set_caption(caption = "Table 2: Top candidates with the highest total votes by year from 1976 to 2020, including party aff
iliation and vote counts.") |>
  theme_vanilla() |>
  autofit()
```

Table 2: Top candidates with the highest total votes by year from 1976 to 2020, including party affiliation and vote counts.

Candidate	Year	Party	Total votes
Biden, Joseph R. Jr	2,020	Democrat	81,268,908
Trump, Donald J.	2,020	Republican	74,216,146
Obama, Barack H.	2,008	Democrat	69,338,846
Obama, Barack H.	2,012	Democrat	65,752,017
Clinton, Hillary	2,016	Democrat	65,677,246
Trump, Donald J.	2,016	Republican	62,692,670
Bush, George W.	2,004	Republican	61,872,711
Mccain, John	2,008	Republican	59,613,835
Romney, Mitt	2,012	Republican	59,379,447
Kerry, John	2,004	Democrat	58,894,554

Voter median income and poverty rate comparison

As we saw on the two previous tables, the most voted parties over the years en every state are the Democrats and Republicans, they have always been on top. Throughout this project, I am going to try finding the relation some demographic factors have in the winning party of each state. For this, first I am going to create a table showing the average voter turnout and poverty rate of each of the two mentioned dominant parties.

```
#creating table
table_dominant_party <- merged_data |>
  group_by(party_simplified) |>
  summarize(
    avg_median_income = round(mean(median_income, na.rm = TRUE), 3),
    avg_poverty_rate = round(mean(prop_poverty, na.rm = TRUE), 3),
    total_states = n()) |>
  ungroup()

#Changing col names
table_dominant_party <- table_dominant_party |>
  rename(
    "Party" = party_simplified,
    "Median Income" = avg_median_income,
    "Poverty" = avg_poverty_rate,
    "Number of states won" = total_states)

#Giving format to table
table_dominant_party |>
  flextable() |>
  set_caption(caption = "Table 3: Average Voter Median Income and Poverty Rate by Dominant Party (2008-2020)") |>
  theme_vanilla() |>
  autofit()
```

Table 3: Average Voter Median Income and Poverty Rate by Dominant Party (2008-2020)

Party	**Median Income**	**Poverty**	**Number of states won**
Democrat	62,746.08	0.122	99
Republican	52,501.54	0.147	101

We can see how median income is higher for democrats, exactly an 19.5% higher. Regarding the poverty rate, the one for republican states is higher by around a 20%. We will later explore this matter further with some graphs.

b. Data Visualizations

For many of the visualizations, we will use the different colors of the main parties:

```
party_colors <- c(
  "Democrat" = "#0015BC",
  "Republican" = "#FF0000",
  "Libertarian" = "#FDBB30",
  "Other" = "grey")
```

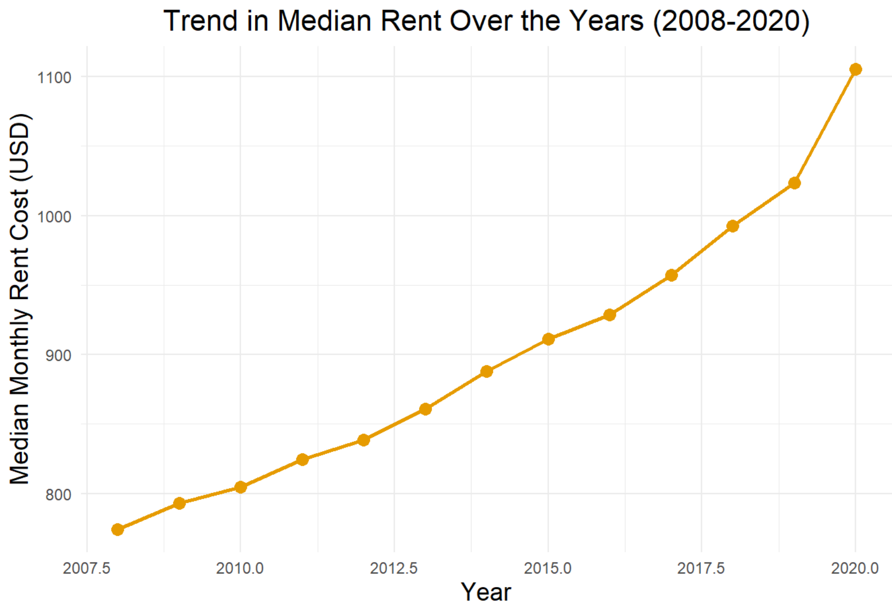
Income over the years

Housing affordability has become a significant issue in the United States, with median rent rising over the past few decades. This upward trend reflects a combination of economic factors, including inflation, increasing demand for housing, and policy decisions at both the state and local levels. Rising rents often place a huge burden on lower and middle-income families, highlighting the importance of housing policy in political discourse.

With some visualizations, we will try to examine how median rent has evolved over time and its connections to political dynamics.

First, I will create a line chart to show the median income from the years 2008-2020:

```
census_data |>
  group_by(year) |>
  summarize(avg_rent = mean(median_monthly_rent_cost, na.rm = TRUE)) |>
  ggplot(aes(x = year, y = avg_rent)) +
  geom_line(size = 1, color = "#E69F00") +
  geom_point(size = 3, color = "#E69F00") +
  labs(
    title = "Trend in Median Rent Over the Years (2008-2020)",
    x = "Year",
    y = "Median Monthly Rent Cost (USD)",
    caption = "Source: tidycensus R package and United States Census website") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16),
     plot.caption = element_text(hjust = 0.5, size = 12, face = "italic"),
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14))
```

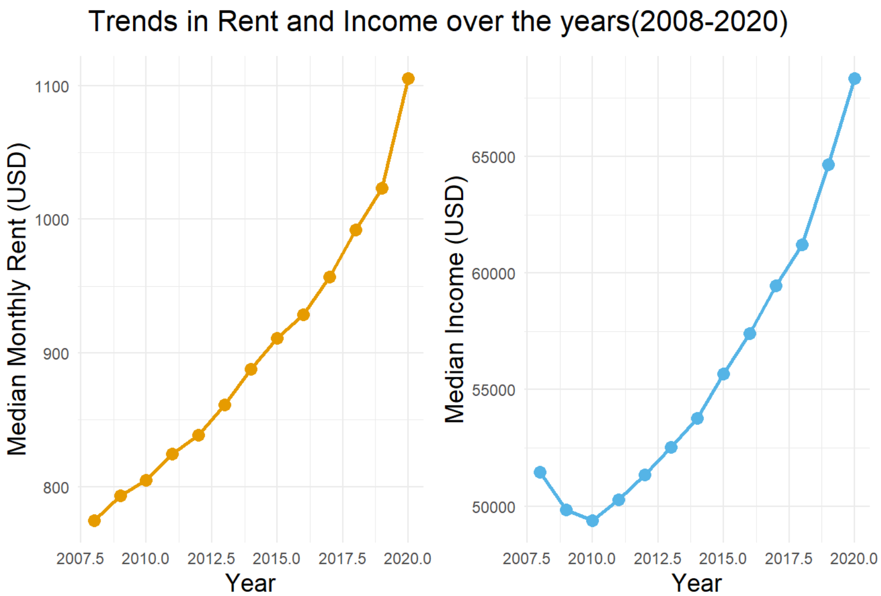


Source: tidycensus R package and United States Census website

We can clearly see the rising pattern stated before.

Next, we will compare the income and monthly rent cost by creating two different graphs for both variables.

```
summary_data <- census_data |>
  group_by(year) |>
  summarize(
    avg_rent = mean(median_monthly_rent_cost, na.rm = TRUE),
    avg_income = mean(median_income, na.rm = TRUE))

# Line Chart for Rent
plot_rent <- ggplot(summary_data, aes(x = year, y = avg_rent)) +
  geom_line(size = 1, color = "#E69F00") +
  geom_point(size = 3, color = "#E69F00") +
  labs(
    x = "Year",
    y = "Median Monthly Rent (USD)") +
  theme_minimal() +
  theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14))

# Line Chart for Income
plot_income <- ggplot(summary_data, aes(x = year, y = avg_income)) +
  geom_line(size = 1, color = "#56B4E9") +
  geom_point(size = 3, color = "#56B4E9") +
  labs(
    x = "Year",
    y = "Median Income (USD)") +
  theme_minimal() +
  theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14))

# Combine the Plots Side by Side
combined_plot <- plot_rent + plot_income +
  plot_layout(widths = c(1.5, 1.5)) +
  plot_annotation(
    title = "Trends in Rent and Income over the years(2008-2020)",
    caption = "Source: tidycensus R package and United States Census website",
    theme = theme(
      plot.title = element_text(hjust = 0.5, size = 16),
      plot.caption = element_text(hjust = 0.5, size = 12, face = "italic")))

print(combined_plot)
```

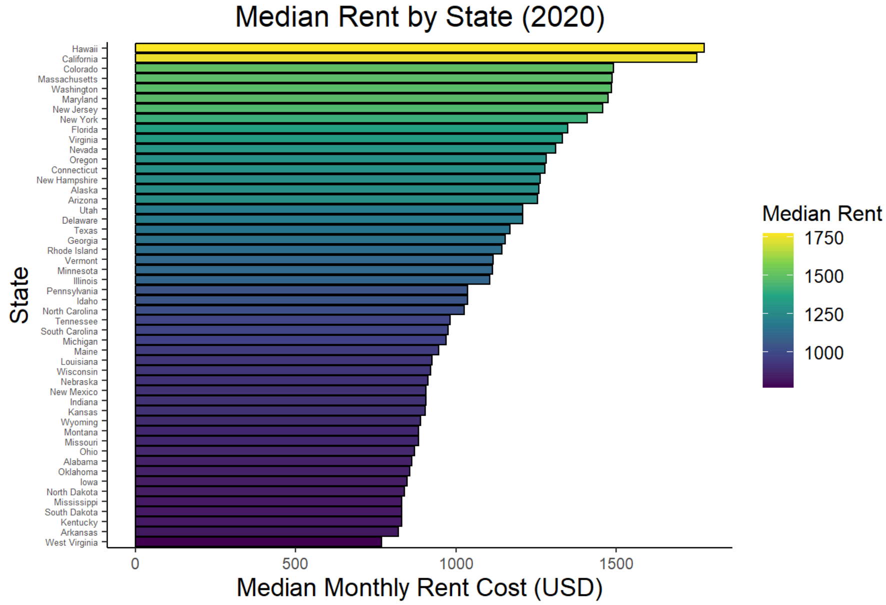


Source: tidycensus R package and United States Census website

In the above two graphs, we can see a similar pattern for rent and income over the years. Both of them have had an even greater increase in the last years, especially from 2018-2020. The only difference we can see is a drop in the median income around the year 2009, which doesn't change the ascending pattern much. This is great for the population, since it means that, even though rent prices are increasing, income is increasing too, so even though rent will be higher, it will still be afordable.

Finally, we will focus on the year 2020. For this year, we will create a bar chart of the median rent by state for this year.

```
merged_data |>
  filter(year == 2020) |>
  ggplot(aes(x = reorder(state, median_monthly_rent_cost), y = median_monthly_rent_cost, fill = median_monthly_rent_cost)) +
  geom_bar(stat = "identity", color = "Black") +
  coord_flip() +
  labs(
    title = "Median Rent by State (2020)",
    caption = "Source: tidycensus R package and United States Census website",
    x = "State",
    y = "Median Monthly Rent Cost (USD)",
    fill = "Median Rent") +
  scale_fill_viridis_c(option = "D") +  # Apply Viridis color scale
  theme_classic() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16),
    plot.caption = element_text(hjust = 0.5, size = 12, face = "italic"),
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    legend.title = element_text(size = 12),
    axis.text.y = element_text(size = 5),
    legend.text = element_text(size = 10))
```



Source: tidycensus R package and United States Census website

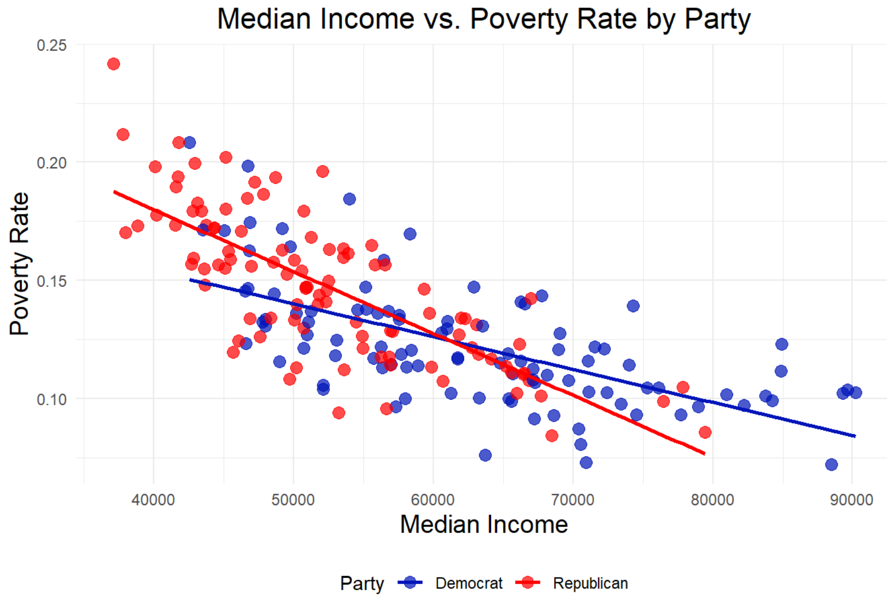
We can see how California and Hawaii have the highest median rent cost, of around 1750. A couple of states have a value above 1500 and most of them are concentrated around 1000 and 1500. The state with the lowest median monthly rent cost is West Virginia.

Median Income vs Poverty Rate by Party

We are now going to further explore if there is an effect by median income and poverty rate on the winning party, while also exploring the relationship between this two factors. We will conduct a graphical analysis by the use of a scatter plot.

```
merged_data |>
  ggplot(aes(x = median_income, y = prop_poverty, color = party_simplified)) +
  geom_point(size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Median Income vs. Poverty Rate by Party",
    caption = "Source: tidycensus R package, United States Census website, MIT Election Data and Science Lab, 2017",
    x = "Median Income",
    y = "Poverty Rate",
    color = "Party") +
  scale_color_manual(values = party_colors) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16),
    plot.caption = element_text(hjust = 0.89, face = "italic"),
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    legend.position = "bottom")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Source: tidycensus R package, United States Census website, MIT Election Data and Science Lab, 2017

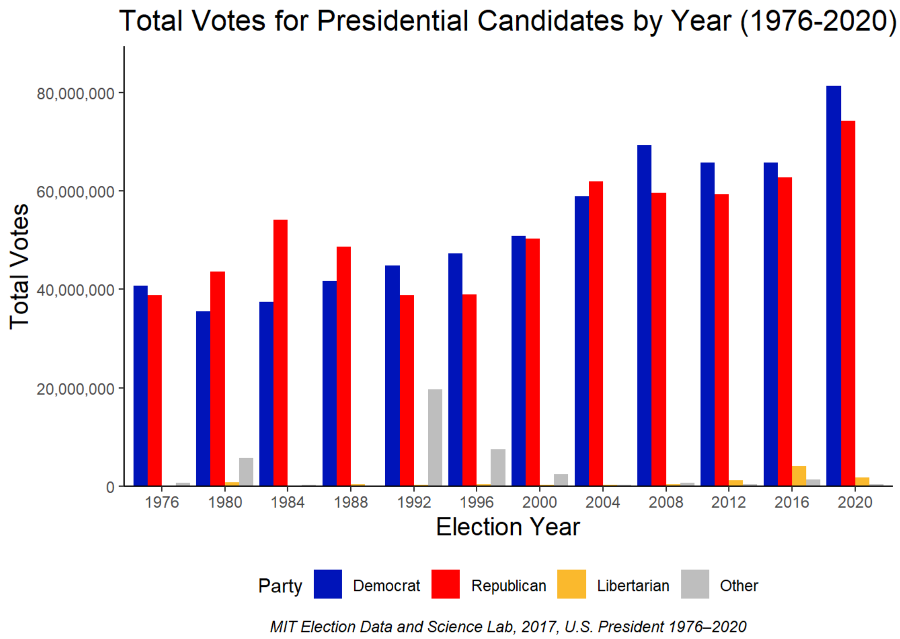
As we can see above in the scatter plot, each point represents one State in one election year. Democrat winning states are colored in blue and Republican states in red. We can also see the two different linear regression or trend lines, one for each party. These lines show the relationship between income and poverty rate.

As expected, the correlation between poverty rate and median income for both parties is negative. This means that, the higher the median income is in a state, the lower its poverty rate is, which makes sense. We can see how the red line is steeper than the blue line. Therefore, median income has a stronger inverse relationship with poverty rate in Republican states. Also, we can clearly appreciate how the red dots are more concentrated on the top left and the blue ones on the bottom right. This means that, by looking at the graph, we could interpret that Republican states have less median income and more poverty rate than Democrat ones. We can also see how Republican states' median income doesn't pass 80000 in any state, and Democrat states' median income doesn't drop bellow 45000.

Elections popular vote by year and party

Now, I am going to create a bar chart that shows total votes for each presidential candidate for each year grouped by party using the variable party_simplified to only show democrats, republicans, libertarians and other as the parties.

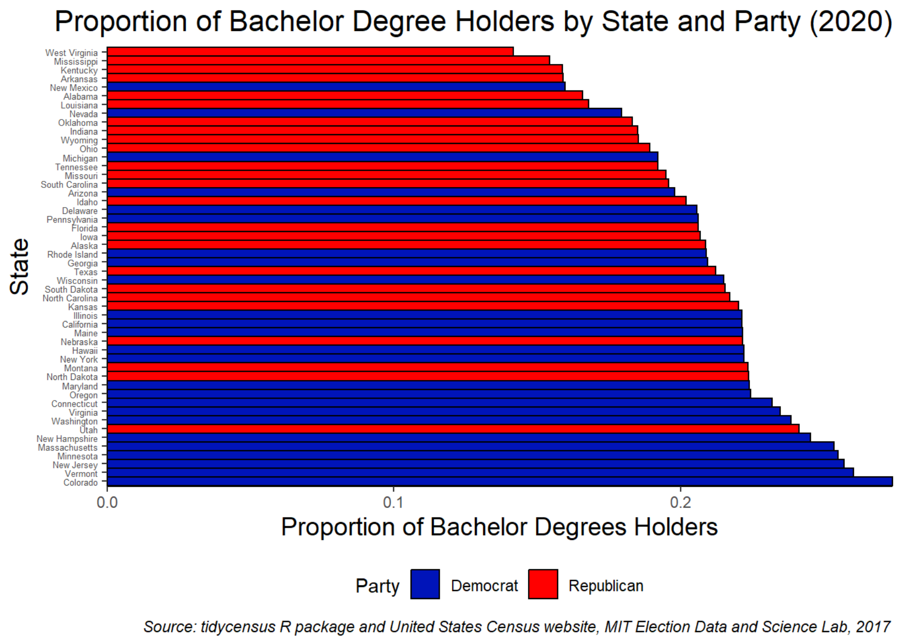
```
# Selecting the order for the bars
elections_new <- elections_new |>
  mutate(party_simplified = factor(party_simplified, levels = c("Democrat", "Republican", "Libertarian", "Other")))

# Creating the plot
elections_new |>
  ggplot(aes(x = factor(year), y = total_votes_us, fill = party_simplified)) +
  geom_col(position = position_dodge(width = 0.9)) +
  labs(
    title = "Total Votes for Presidential Candidates by Year (1976-2020)",
    x = "Election Year",
    y = "Total Votes",
    caption = "MIT Election Data and Science Lab, 2017, U.S. President 1976-2020",
    fill = "Party") +
  scale_fill_manual(values = party_colors) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1)), labels = scales::comma) +
  theme_classic() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16),
    plot.caption = element_text(hjust = 0.5, face = "italic"),
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    legend.position = "bottom")
```



We can see how, over the years, Republicans and Democrats have lead the vote count.In the last 4 elections, the democrats have won the popular vote every time, although this doesn't mean they won the election. As we now, the electoral college determines who the president is, so, for example, in the 2016 elections, even though they got less votes, the Republicans won the elections and Trump became president. The same thing happened in the year 2000.

Proportion of Bachelor Degree Holders by State

```
merged_data |>
  filter(year == 2020) |>
  ggplot(aes(x = reorder(state, -prop_bachelors), y = prop_bachelors, fill = party_simplified)) +
  geom_bar(stat = "identity", width = 1, color = "Black") +
  coord_flip() +
  labs(
    title = "Proportion of Bachelor Degree Holders by State and Party (2020)",
    caption = "Source: tidycensus R package and United States Census website, MIT Election Data and Science Lab, 2017",
    x = "State",
    y = "Proportion of Bachelor Degrees Holders",
    fill = "Party") +
  scale_fill_manual(values = party_colors) +
  scale_y_continuous(expand = c(0, 0)) +
  theme_classic() +
  theme(
    plot.title = element_text(hjust = 1, size = 16),
    plot.caption = element_text(hjust = 0.95, face = "italic"),
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    axis.text.y = element_text(size = 5),
    legend.position = "bottom")
```



On the above graph, we can start to see a pattern. Republican states are concentrated at the top, with a smaller proportion of bachelor degree holders. As we look at the bottom of the graph, we can start to see more blue or Democrat states, meaning they have a higher proportion of bachelor degrees.

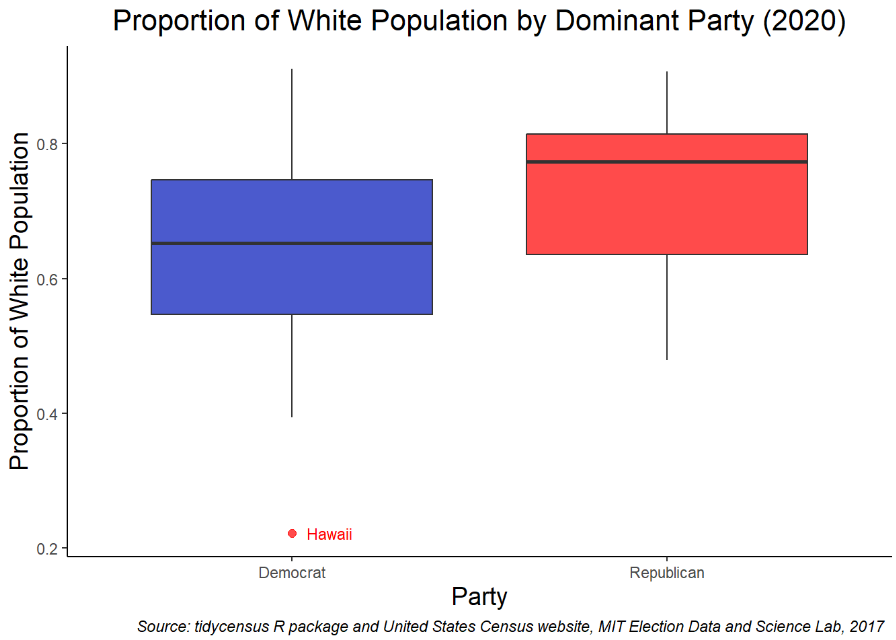
Proportion of White population by party

The topic of racial dynamics in the United States has always been central to its political discourse, and it gained particular intensity during the presidency of Donald Trump. His administration often faced criticism for policies that were viewed as catering to the concerns of majority White populations, while failing to address racial inequalities in the country.

We are now going to focus on the proportion of White populations across states dominated by Democrats and Republicans in the 2020 election. By comparing these proportions, this graph seeks to provide insights into the racial composition of states and how it correlates with political dominance, shedding light on the demographic divides that continue to shape the U.S. political landscape.

This analysis will be conducted by the use of two boxplots, one for each party, showing the proportion of white population in each state for the year 2020.

```
ggplot() +
  geom_boxplot(
    data = merged_data |> filter(year == 2020),
    aes(x = party_simplified, y = prop_white, fill = party_simplified),
    outlier.color = "red", outlier.size = 2, alpha = 0.7) +
  geom_text(
    data = merged_data |> filter(year == 2020, state == "Hawaii"),
    aes(x = party_simplified, y = prop_white, label = state),
    nudge_x = 0.1,  # Adjust the horizontal position of the label as needed
    size = 3,
    color = "red") +
  labs(
    title = "Proportion of White Population by Dominant Party (2020)",
    caption = "Source: tidycensus R package and United States Census website, MIT Election Data and Science Lab, 2017",
    x = "Party",
    y = "Proportion of White Population",
    fill = "Party") +
  scale_fill_manual(values = party_colors) +
  theme_classic() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16),
    plot.caption = element_text(hjust = 0.9, face = "italic"),
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    legend.position = "none")
```



Above we can see the two boxplots, each with its party's color. It is clear that the mean and median of the Republican-dominated states white population proportion is higher than the Democrat states one. The mean for Democrat states' white population proportion is 0.64, while for Republican ones it is 0.73. This suggests that states with larger white populations may have leaned toward Republican dominance. We need to keep in mind that, to correctly interpret these results and draw conclusions, we would need to consider additional factors. It is important to highlight that we have one outlier in the Democrat boxplot. It corresponds to Hawaii, and can be attributed to its unique geographic location, history and demographic makeup.

4. Monte Carlo Methods of Inference

Now, we will further explore the relationship between Bachelor degree holders and the dominant party for each state for the year 2020. We want to determine if the proportion of individuals with a bachelor's degree is statistically significantly higher in states where Democrats are the dominant party. To do this, we will use a permutation test.

The hypothesis are:

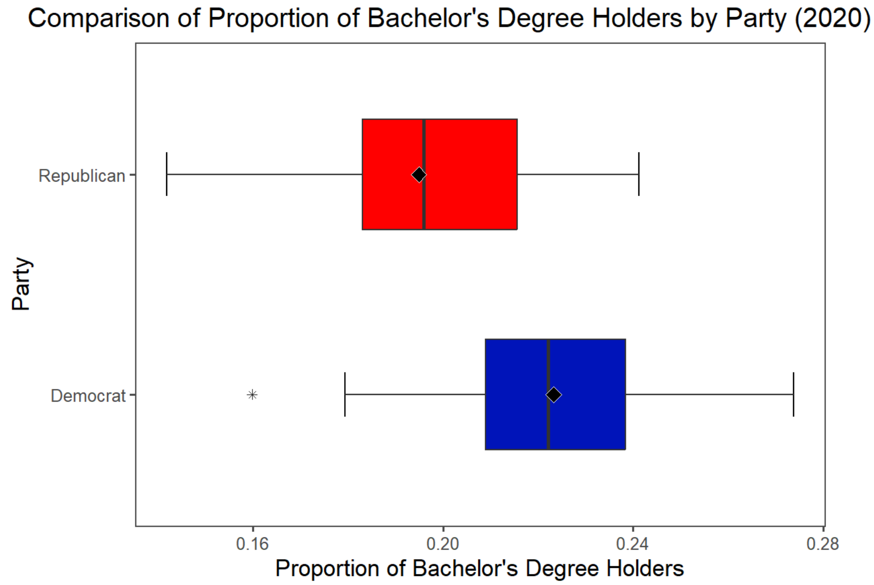
H_0: $\mu_1 = \mu_2$

H_a: $\mu_1 > \mu_2$

Where μ_1 represents the mean proportion of individuals with a bachelor's degree in Democrat-dominated states, and μ_2 represents the mean proportion of individuals with a bachelor's degree in Republican-dominated states

Let's compare both proportions:

```
# Filter data for the year 2020
data_2020 <- merged_data |> filter(year == 2020)

#Boxplots:
ggplot(data_2020, aes(x = party_simplified, y = prop_bachelors, fill = party_simplified)) +
  stat_boxplot(geom = "errorbar", width = 0.2, coef = 1.5) +
  stat_boxplot(geom = "boxplot", width = 0.5, coef = 1.5, outlier.shape = 8) +
  stat_summary(fun = "mean", geom = "point", shape = 23, fill = "black", color = "white", size = 3) +  # Mean points
   scale_fill_manual(values = party_colors) +
  coord_flip() +
  labs(
    y = "Proportion of Bachelor's Degree Holders",
    x = "Party",
    title = "Comparison of Proportion of Bachelor's Degree Holders by Party (2020)",
    fill = "Party") +
  theme_few() +
  theme(
    plot.title = element_text(hjust = 0.7, size = 15, ),
    axis.title.x = element_text(size = 13),
    axis.title.y = element_text(size = 13),
    plot.margin = margin(t = 20, r = 30, b = 20, l = 20),
    legend.position = "none")
```

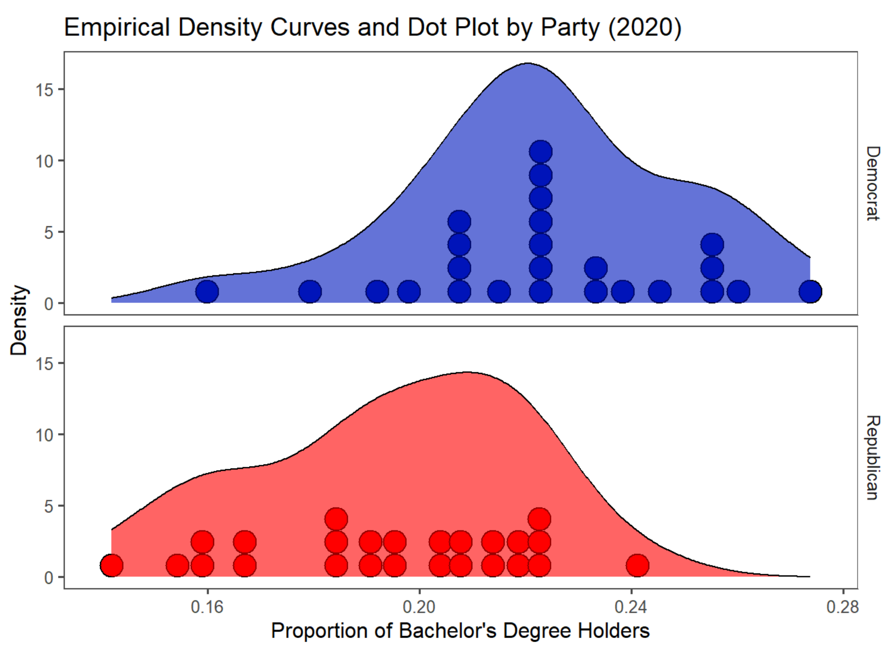


We can visually see how proportion of Bachelor Degree Holders is higher for Democrat dominated states.

Let’s now look at the empirical density curves:

```
ggplot(data_2020, aes(x = prop_bachelors, fill = party_simplified)) +
  geom_dotplot() +
  geom_density(color = "black", alpha = 0.6) +
  scale_fill_manual(values = party_colors) +
  facet_grid(party_simplified ~ .) +
  labs(
    x = "Proportion of Bachelor's Degree Holders",
    title = "Empirical Density Curves and Dot Plot by Party (2020)",
    y = "Density") +
  theme_few() +
  theme(
    legend.position = "none",)
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## `binwidth`.
```



Let’s implement the two sample t-test

```
# Relevel party_simplified to make "Democrat" group 1
data_2020 <- data_2020 |>
  mutate(party_simplified = fct_relevel(party_simplified, "Democrat", "Republican"))

# Implementing two-sample t-test
ttest_result <- t.test(formula = prop_bachelors ~ party_simplified,
                       alternative = "greater",
                       data = data_2020)
```

Displaying the results:

```
# Tidy the t-test result and format it for display
ttest_result |>
  broom::tidy() |>
  flextable() |>
  colformat_double(digits = 3) |>
  set_formatter(p.value = function(x) {format.pval(x, digits = 3)}) |>
  set_caption("Results of two-sample t-test comparing proportion of bachelor's degree holders between Democrat and Republica
n states.") |>
  autofit() |>
  fit_to_width(max_width = 7)
```

Results of two-sample t-test comparing proportion of bachelor's degree holders between Democrat and Republican states.

estimate	estimate1	estimate2	statistic	p.value	parameter	conf.low	conf.high	method	alternative
0.028	0.223	0.195	3.859	0.00017	47.978	0.016	Inf	Welch Two Sample t-test	greater

Now, let's do the randomization test, which requires equal variances.

```
data_2020 |>
  group_by(party_simplified) |>
  summarize(
    Mean = mean(prop_bachelors, na.rm = TRUE),
    n = n(),
    SD = sd(prop_bachelors, na.rm = TRUE),
    Variance = var(prop_bachelors, na.rm = TRUE))  |>
  flextable() |>
  colformat_double(digits = 3) |>
  autofit()
```

party_simplified	Mean	n	SD	Variance
Democrat	0.223	25	0.026	0.001
Republican	0.195	25	0.026	0.001

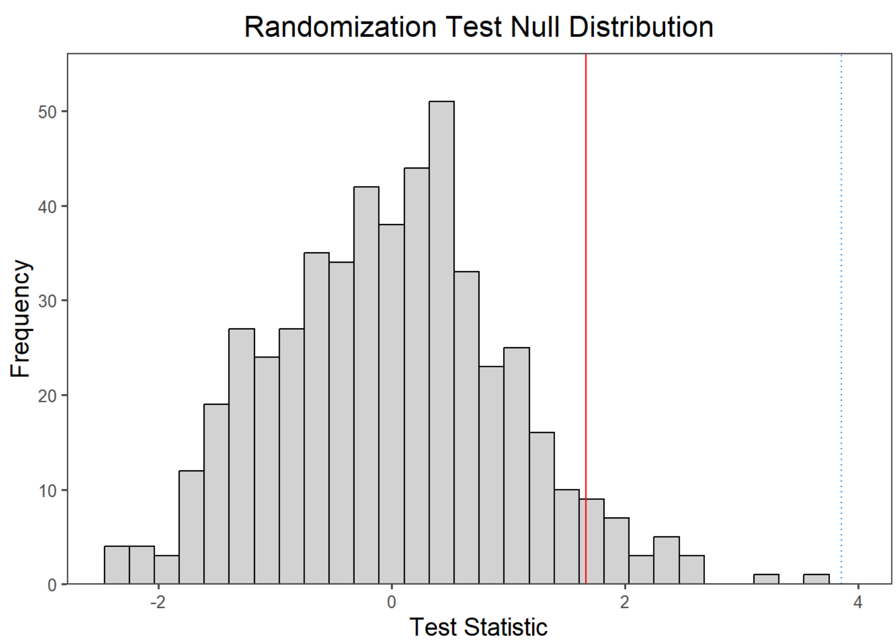
We can see that there are equal variances. Let's implement the two-sample t-test for 500 random permutations of individuals to create our null distribution of test statistics for the randomization test.

```
# Number of permutations to do
n_permutations <- 500

# Instantiating vector for test statistics
permutation_statistics <- vector(length = n_permutations)

# Calculating t-test statistic for each permutation
for (p in 1:n_permutations) {
  permutation_statistics[p] <- t.test(
    formula = prop_bachelors ~ party_simplified,
    alternative = "greater",
    data = data_2020 |>
      mutate(party_simplified = sample(party_simplified, replace = FALSE))) |>
    broom::tidy() |>
    pull(statistic)
}
```

Let's create a histogram displaying the null distribution obtained for the randomization test. Vertical lines will be added to the plot to indicate where the 5th percentile is (a solid red line), and where our observed test statistic is (dotted blue line).

```
# Tidying into a tibble
permutation_results <- tibble(Value = permutation_statistics)

# Visualizing the randomization test null distribution
permutation_results |>
  ggplot(aes(x = Value)) +
  geom_histogram(color = "black", fill = "#D3D3D3") +
  geom_vline(xintercept = quantile(permutation_statistics, probs = 0.95),
             color = "red") +
  geom_vline(xintercept = ttest_result$statistic,
             color = "dodgerblue",
             linetype = "dotted") +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Randomization Test Null Distribution",
    y = "Frequency",
    x = "Test Statistic") +
  theme_few() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16),
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14))
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



ON the graph, we can clearly see how our observed test statistic line (blue) has a higher value than the 5th percentile (red line) Let's now calculate the p-value for the randomization test:

```
# Calculating the randomization test p-value
mean(permutation_statistics >= ttest_result$statistic)
```

```
## [1] 0
```

Do we reject or fail to reject the null at the 5% significance level?

Decision: We reject the null hypothesis since $0 < \alpha = 0.05$

Interpretation in context: We have sufficient evidence at the 5% significance level that the average proportion of bachelor degree holders in Democrat-dominated states is greater than in Republican-dominated states.

5. Conclusions / Main Takeaways

This analysis explored the last election results and candidates, different demographic factors, and the relationship between them. The objective was to uncover meaningful insights and evaluate the significance of observed trends. The visualizations highlighted noticeable differences in educational attainment between Democrat- and Republican-dominated states, with Democrat-dominated states exhibiting a higher average proportion of bachelor's degree holders. Also, some other demographics relationships were showed, such as the one between rent cost and income or between income and poverty, which is a negative relation. It is also important to highlight that white people proportion is higher in

Republican-dominated states. Regarding politics, the dominance of the two big parties, Democrats and Republicans is huge, and there is a clear increasing pattern in the number of votes in each election, which can be attributed to population growth but also to higher voter turnout. This project demonstrates the importance of integrating robust statistical methods with clear visualization and thoughtful interpretation.